Parsed, Plausible, Wrong: Output Format Controls Silent Errors in Small-Model Semantic Parsing

Umar Ahmed and Maxim Duanev and Mohammad Nour Al Awad and Sergey Ivanov and Ivan V. Khodnenko
ITMO University
{uakhmed,mduanev,svivanov}@itmo.ru, MohammadNourAlAwad@itmo.ru, ivan.khodnenko@itmo.ru

Abstract

A small fine-tuned parser that emits a well-formed but wrong tree fails *silently*: nothing downstream can tell. We ask whether the output format, a free design choice, controls how often this happens. On the PIZZA ordering benchmark we fine-tune two sub-billion models (Qwen3.5-0.8B and LFM2.5-350M) with LoRA on six lossless encodings of the same target tree, from 44.3 down to 10.9 tokens per order, including two matched-length pairs that vary surface form at fixed length. Across 300 runs (5 training pool sizes, 5 seeds) with a hierarchical bootstrap and Holm correction fixed before the results were read, shortening alone never helps, but writing the target in ordinary words does. The word encoding is the most accurate format on both models (87.5% vs. 85.8% exact match for the native s-expression on Qwen) while emitting 4.1× fewer tokens, and it cuts the silent error rate by 2.3× on Qwen and 1.8× on LFM2, separating at every pool size. Error analysis locates much of the gain in negation. All runs also learn to refuse off-menu input (held out: 95.9–99.0% handled) and to apply in-utterance self-corrections. Strict refusal has a measured price: 2.1–10.1% of real, mostly polite, orders are refused.

1 Introduction

Small language models are attractive for narrow parsing tasks: they fine-tune in minutes, run on a laptop or a tablet, and cost almost nothing per call. Their weakness is reliability (Ji et al., 2023), and in structured prediction an error comes in two kinds. An output that does not parse, or that the model marks as a refusal, is *detectable*: the caller rejects it and asks again. An output that parses into a plausible tree that is not what the customer said is *silent*: the order goes through, wrong. This paper treats the silent error rate S as the quantity a deployment should minimise, and reports exact match A beside it for comparison with prior work (Section 5).

The work began with a deployed café ordering parser whose compact output format was chosen for speed; its small hand-written test set could not tell formats apart, so we moved to a public benchmark with a human-written test set. Our question is whether the target format is itself a lever on silent errors. The same tree can be written as an s-expression, as JSON, as `key=value` pairs, as a terse symbolic language or as plain words, and the choice changes how many tokens the model emits and how much of its pretraining it can reuse. The intuitive hypothesis, that fewer output tokens leave less room for error, turns out to be wrong in the form we first stated it. Shortening helps only when the surface stays in natural words.

Contributions. (1) A ladder of six lossless encodings of the PIZZA target tree (Arkoudas et al., 2022), with two matched-length pairs that separate length from surface form (Section 4). (2) Evidence across two model families, five pool sizes and five seeds that length alone is not the mechanism: the word encoding beats its symbolic twin of equal length in 5+5 of 10 model×size cells, separating in 5+4 after Holm correction, and on LFM2 it wins while costing *more* tokens (Section 6.2). (3) The silent error rate as the deployment objective, inside a four-way split of every test set. The word encoding has the lowest S on both models, and an error analysis shows where the difference comes from (Section 6.3). (4) A guardrail trained into every run, refusing off-menu input and applying spoken self-corrections, with its benefit and its cost measured on held-out data (Section 6.4).

2 Related Work

Target representation. That the meaning representation affects a parser is long established: Dong and Lapata (2016) decode logical forms as trees rather than sequences, and seq2seq parsers for task-oriented dialogue generate the target directly (Ron-

gali et al., 2020). Closest to us, Shin et al. (2021) show that a frozen language model prompted to produce a canonical natural-language form, mapped to the representation by a grammar, beats prompting for the representation itself. We find the same phenomenon under fine-tuning, where the model is free to learn any notation, and we separate naturalness from length with matched pairs and two vocabularies.

Format restrictions and constrained decoding. Tam et al. (2024) find that forcing an LLM into JSON costs reasoning accuracy. Constrained decoding makes invalid output impossible (Scholak et al., 2021; Geng et al., 2023; Willard and Louf, 2023); it can remove our unparseable outputs, but it cannot make a well-formed tree correct, so the silent rate is the part of the problem it does not reach.

Abstention and self-correction. Knowing when not to answer has been studied for reading comprehension (Rajpurkar et al., 2018) and under domain shift (Kamath et al., 2020). Spoken self-repair is a classic topic in disfluency research (Shriberg, 1994), and Disfl-QA shows that disfluent input breaks QA models (Gupta et al., 2021). Task-oriented parsing datasets such as TOP, TOPv2 and MTOP (Gupta et al., 2018; Chen et al., 2020; Li et al., 2021), and PIZZA itself, contain neither off-domain refusals nor self-corrections; we add both as synthetic training data and evaluate them on held-out vocabulary and cue phrases.

Tokenization and evaluation. Subword segmentation (Sennrich et al., 2016) sets what a string costs, and vocabulary fit moves accuracy (Rust et al., 2021). We follow recommendations to report seed variance and significance (Dror et al., 2018; Bouthillier et al., 2021), using a bootstrap (Efron and Tibshirani, 1993) with Holm's correction (Holm, 1979).

3 Task and Data

PIZZA (Arkoudas et al., 2022) maps spoken pizza and drink orders to trees with an unordered exact-match criterion. Its training pool of 2,456,446 utterances is generated from a grammar; the 348-utterance development set and the 1,357-utterance test set are written by people, which makes the test set the strongest evidence in this paper. The pool contains 3 rows identical to a development or test utterance. We drop them from training, and verify on every training subset we draw, at every size and seed, that no evaluation utterance remains, exactly or after normalisation. The splits are used as published.

Format	Tok.	Target
EXR	38	`(ORDER (PIZZAORDER (NUMBER 1 ) (COM...`
JSON	33	`{"orders":[{"type":"p","attrs":[{"n...`
EXR-SHORT	25	`(O (P (N 1) (C (Q EXTRA) (T CHEESE)...`
TAGGED	13	`pizza n=1 extra=cheese topp=pepperoni`
DSL	8	`p 1 +cheese pepperoni`
WORDS	7	`pizza 1 extra cheese pepperoni`

Table 1: One test order, *"i want a pie with pepperoni and extra cheese"*, in the six encodings (long targets truncated). Tokens are Qwen's.

Refusals. PIZZA has no notion of "not an order": a model trained on it turns *"hehe hehe"* into a pizza. We add generated negatives making up 30% of every training set: chatter, off-menu requests, gibberish, long text, and real orders with an off-menu clause or an unknown word inserted. The rule is strict: anything that is not purely an order is refused whole. Each format has one short refusal string that cannot parse as an order in that format (`?` for the three word-like formats, `(REJECT )`, `(R)` and `{"reject":true}` for the others), so all formats train on identical inputs. Vocabulary is split: the 614 evaluation negatives use words never seen in training.

Self-corrections. PIZZA has no self-corrections either, so *"a pizza with cheese, wait no cheese, only tomato sauce"* comes out with both cheese and no cheese. We take real training orders, edit their tree, and append the spoken correction in nine kinds (swap, retract, only, size, number, drink, cancel, restart, and decoys in which a cue word changes nothing), adding 10% of the real orders. A quarter of the cue phrases are held out: the 600 evaluation corrections are test orders with held-out cues. A fixed set of 40 hand-written probes includes the cheese example above, which a user reported against an earlier model. Dev and test remain pure PIZZA, so A and S keep their meaning.

4 Six Encodings of One Tree

Comparing unrelated formats confounds length with form, so we built six encodings of the *identical* tree that decode back to it on every dev and test utterance and on a 50,000-row training sample (Table 1). EXR is the dataset's s-expression. JSON is a conventional encoding of about the same

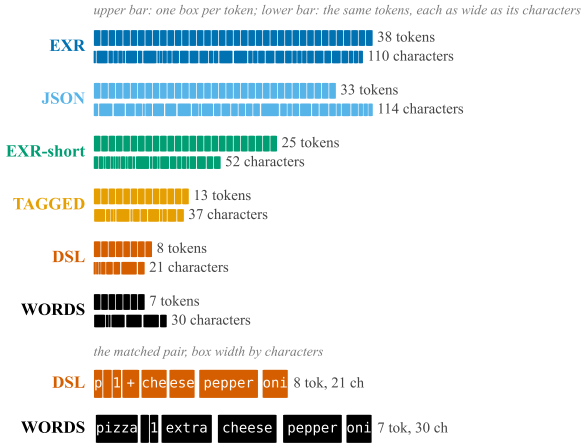


Figure 1: The order of Table 1 under Qwen's tokenizer. For each format the upper bar has one box per token and the lower bar draws the same tokens as wide as their characters (both bars to a common scale across formats). Bottom: the matched pair with its tokens; WORDS is longer as text and shorter in tokens. Boundaries come from the tokenizer's own merge table.

length: the matched-length control for EXR. EXR-SHORT renames EXR's constructors to single letters. TAGGED is flat `key=value` pairs. DSL is a terse grammar with sigils (`+` for an extra quantity, `!` for negation), and WORDS has the same structure as DSL with the markers written as ordinary words; it is DSL's matched-length control. EXR, EXR-SHORT, TAGGED and DSL form a length ladder.

Characters are not tokens. Over the 1,357 gold test trees, WORDS is the longer of the matched pair in characters (43.1 vs. 34.9) but costs fewer Qwen tokens (10.3 vs. 10.8) and about the same number of LFM2 tokens (11.9 vs. 11.7; Appendix Table 7). Figure 1 shows why: `+cheese` splits into three subwords where `extra cheese` stays two. Symbols save characters, not tokens. LFM2's vocabulary is smaller and inflates the punctuation-heavy formats most (JSON 45.6→64.7 tokens, WORDS 10.3→11.9).

5 Setup and Statistics

Models and training. Qwen3.5-0.8B (Qwen Team, 2025) and LFM2.5-350M (Amini et al., 2025), a largely convolutional model with a vocabulary about four times smaller. LoRA (Hu et al., 2022) with rank 16 and α=16 on all linear projections, batch 16, cosine schedule, loss on the completion only. Rank is held fixed across formats because the comparison is between formats, not adapters; Hu et al. (2022) found far lower ranks sufficient on their tasks. Training runs to at most 4,000 steps (64,000 examples) with early stopping every 100 steps (patience 10) on the mean of three dev scores: exact match, refusal of dev negatives, and exact match on dev corrections. The learning rate is chosen per model and format from $\{10^{-4}, 2{\cdot}10^{-4}, 5{\cdot}10^{-4}\}$ by dev exact match at a pool of 100k over 3 seeds, then frozen (Appendix Table 8). The token cap per format is 1.25 times its longest target, so no correct output can be truncated.

Grid. Six formats $\times$ pool sizes {1k, 10k, 100k, 1M, 2,456,443} $\times$ 5 seeds $\times$ two models, 300 runs, every one with the refusal and correction data. A seed fixes the training subset, the negatives and the corrections, so formats are paired by seed. Since no run sees more than 64,000 examples, the larger sizes differ in the diversity of the pool a run samples from, not in examples seen; we call the axis *pool size*.

Metrics. Every test order ends in exactly one of four outcomes:

$$\underbrace{A}_{\text{correct}} + \underbrace{S}_{\text{silent}} + \underbrace{D_{\text{parse}}}_{\text{unparseable}} + \underbrace{D_{\text{refuse}}}_{\text{refused}} = 1.$$

A is exact match under the dataset's own matcher. S is output that parses into a tree that does not match: the caller cannot detect it. D_{parse} does not parse, and D_{refuse} is a refusal of a real order; both are visible to the caller. We read them in this order: S is the deployment objective, because those errors reach the customer; A is reported beside it for comparison with prior work; and D_{refuse} is the price of the guardrail. Every table shows A and S together.

Statistics. Fixed before any result was read. A cell is the mean of 5 seeds. Intervals and p-values come from a hierarchical bootstrap (10,000 resamples) over seeds (paired by seed index across formats) and test utterances (paired across formats). A comparison *separates* when its Holm-adjusted $p < 0.05$, the family being the five pool sizes of one comparison on one model. Retraining one cell twice moved test exact match by 1.5 points on Qwen and 4.9 on LFM2, against a seed SD in that cell of 1.9 and 3.0; we therefore use the seed spread and the bootstrap, not a single rerun, as the measure of noise.

Format	Tok.	A (EM) ↑	S (silent) ↓	D_{parse}	D_{refuse}	Inv./utt.	ms
Qwen3.5-0.8B							
EXR	44.3	85.8 ±1.6	10.3 ±0.8	0.6 ±0.8	3.3 ±1.1	0.100	1171
JSON	44.1	81.6 ±1.3	14.9 ±1.3	0.3 ±0.3	3.2 ±1.0	0.126	1220
EXR-SHORT	32.3	75.2 ±3.6	17.4 ±4.1	3.3 ±2.0	4.2 ±2.2	0.140	903
TAGGED	22.2	85.9 ±2.9	11.6 ±2.4	0.5 ±0.3	2.1 ±0.8	0.095	616
DSL	11.3	79.7 ±1.9	8.3 ±2.3	6.7 ±3.2	5.4 ±2.3	0.070	325
WORDS	10.9	**87.5** ±0.8	**4.5** ±1.5	4.6 ±0.8	3.4 ±1.8	0.018	317
LFM2.5-350M							
EXR	52.4	74.0 ±2.5	17.9 ±2.8	0.2 ±0.2	7.8 ±4.2	0.123	821
JSON	57.8	70.7 ±3.6	19.1 ±1.2	0.1 ±0.2	10.1 ±3.3	0.135	926
EXR-SHORT	35.6	70.8 ±0.7	16.9 ±2.6	2.1 ±1.7	10.1 ±2.3	0.141	571
TAGGED	24.4	71.6 ±2.5	21.4 ±3.8	0.7 ±0.4	6.3 ±3.9	0.150	395
DSL	11.4	72.0 ±3.0	13.1 ±2.6	5.7 ±1.5	9.2 ±3.5	0.086	178
WORDS	12.0	**77.4** ±3.2	**10.0** ±1.6	5.8 ±0.8	6.7 ±3.0	0.051	186

Table 2: Test results at the full pool (2,456,443), mean ± seed SD over 5 seeds, in percent of the 1,357 human-written test orders. $A + S + D_{\text{parse}} + D_{\text{refuse}} = 100$. Bold: best format in the column; underlined: not separable from the best (Holm, $p \geq 0.05$). Tok.: mean output tokens. Inv./utt.: invented entities per order. ms: median latency at batch size one on one GPU, mean over seeds.

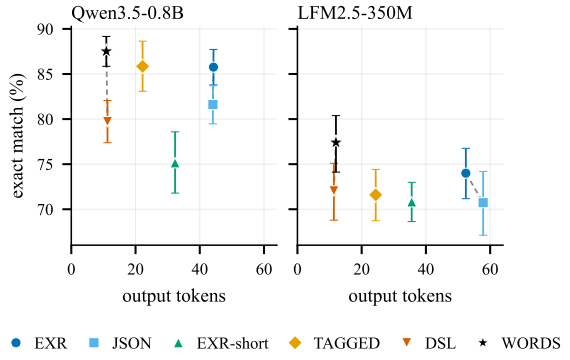


Figure 2: Exact match against output tokens at the full pool, 95% bootstrap intervals. Dashed lines join the matched-length pairs (EXR/JSON, DSL/WORDS).

Difference (points)	1k	10k	100k	1M	2.5M
Qwen3.5-0.8B					
WORDS−DSL	+6.7†	+4.8†	+6.2†	+5.8†	+7.8†
EXR−JSON	+9.3†	+3.8	+3.6	+5.5†	+4.2†
EXR−EXR-SHORT	+6.2†	+11.6†	+10.3†	+9.3†	+10.6†
WORDS−EXR	+4.0	+1.8	+1.6	+0.0	+1.8
S: EXR−WORDS	+11.4†	+5.1†	+4.4†	+5.0†	+5.8†
LFM2.5-350M					
WORDS−DSL	+8.9†	+6.7†	+4.3†	+4.3	+5.4†
EXR−JSON	+3.4	+1.5	+3.4	+1.5	+3.3
EXR−EXR-SHORT	-0.9	+2.3	+1.4	+1.1	+3.2†
WORDS−EXR	+7.2†	+4.9†	+3.5	+4.4†	+3.4
S: EXR−WORDS	+11.2†	+8.6†	+7.7†	+8.7†	+7.9†

Table 3: Paired differences in exact match (points; last row: silent rate, positive = WORDS lower) at every pool size, both models. †: separates (Holm within the row). 95% intervals in Appendix Table 6.

6 Results

Table 2 gives the full-pool results. On both models WORDS has the highest exact match (87.5% on Qwen; EXR, TAGGED not separable from it) and the lowest silent rate (no other format separable on Qwen; DSL on LFM2), with the fewest tokens on Qwen and nearly the fewest on LFM2. For reference, the published BART-Large EXR system reaches 78.6% on this test set with the full pool (Arkoudas et al., 2022); it is a fully fine-tuned encoder-decoder trained without refusals or corrections, so we do not compare further.

6.1 Shorter is not better

Regressing exact match on output tokens across the four ladder formats gives a positive slope (removing tokens lowers accuracy) in 10 of 10 model×size cells, never more than 1.1 points per ten tokens, and it separates in only 1. The ladder is not monotone (Figure 2): EXR-SHORT, which only renames EXR's constructors, loses +10.6 points on Qwen, separating at every size, and +3.2 on LFM2; TAGGED, half as long as EXR, matches it on Qwen. Length does not predict accuracy in either direction.

6.2 Surface form, not length

The matched pairs separate the two explanations (Table 3). WORDS and DSL share their structure and differ by less than a token; they differ only in whether a marker is `+cheese` or `extra cheese`. WORDS is ahead in every one of the 10 cells, by +7.8 points on Qwen [+6.1, +9.5] and +5.4 on LFM2 at the full pool, and separates in all but one. On LFM2 the pair's token order inverts: WORDS

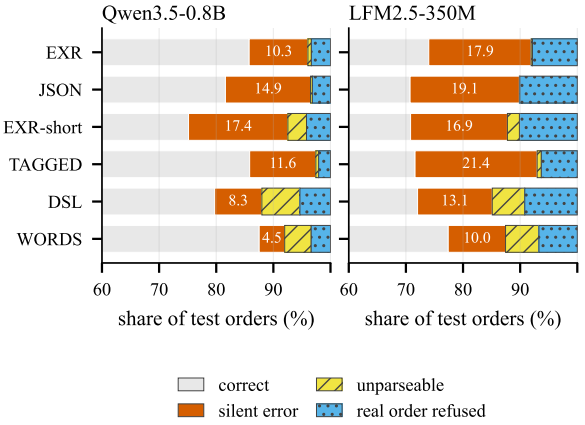


Figure 3: Where the test orders go at the full pool (axis starts at 60%). Labels: silent error rate S.

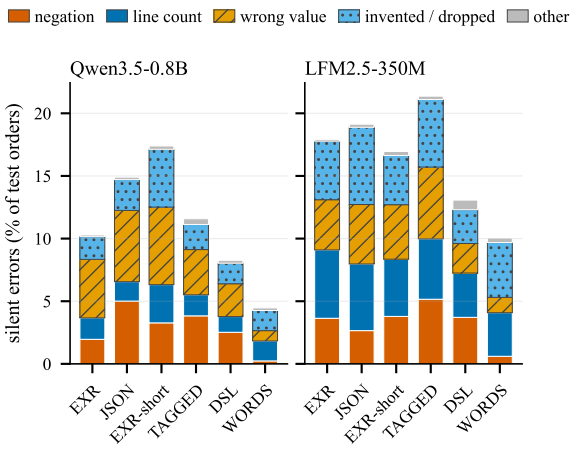


Figure 4: Silent errors by kind at the full pool, as a share of all test orders (bar height = S). Negation: a `NOT` attribute added, lost or misplaced; line count: wrong number of pizzas or drinks.

emits 12.0 tokens against DSL's 11.4 and is still the more accurate. A format that is longer and more accurate is not winning by being short.

EXR and JSON are both bracket-heavy notations, so this pair varies little along the natural-word axis. EXR is ahead in all 10 cells but separates in only 3 (all on Qwen): a consistent but small effect, as expected if what matters is words rather than brackets. Against the native format, WORDS is ahead on both models (+1.8 and +3.4 points at the full pool), separating in some sizes on LFM2 and none on Qwen; we read this as matching EXR's accuracy with 4.1× fewer tokens, not as beating it.

6.3 Silent errors and where they come from

Exact match and silent rate rank formats differently (Figure 3). EXR and JSON almost always parse, so nearly everything they get wrong they get wrong silently. WORDS and DSL fail to parse more often, and those failures are visible. WORDS has the lowest S: 4.5% against 10.3% for EXR on Qwen, a ratio of 2.3 [1.8, 3.1], and 10.0% against 17.9% on LFM2, ratio 1.8 [1.4, 2.3]. The difference separates at every pool size on both models (Table 3, last rows). WORDS also invents fewer entities that the customer never said: 5.5× fewer than EXR on Qwen and 2.4× on LFM2.

Classifying every silent error by what it changes (Figure 4; Appendix Table 11) shows where the difference comes from. Negation is 5% of WORDS's silent errors on Qwen and 6% on LFM2, against 19–34% and 14–28% for the other formats; as a rate, Qwen makes a silent negation error on 2.0% of orders with EXR and 0.2% with WORDS. WORDS writes `no onions` the way customers say it, where the other formats translate it into a `NOT`-construct or a sigil. What WORDS still gets wrong is mostly the number of lines (36% of its silent errors on Qwen).

6.4 The guardrail: refusing and listening

Refusal works for every format. Held-out negatives, built from vocabulary never seen in training, are refused 95.9–99.0% of the time on Qwen and 97.2–98.3% on LFM2 (Table 4). Chatter, gibberish and long text are refused essentially always; the hardest kind is a real order with one unknown word, refused 86–96% of the time on Qwen (Appendix Table 9). Format barely matters here.

The price is refusing real orders. 2.1–5.4% of real test orders are refused on Qwen and 6.3–10.1% on LFM2. Reading them, they are genuine orders in human phrasing the synthetic grammar never produces: greetings, thanks, and stories (*"i'm having a bunch of people over tonight so i need five pizzas…"*). Almost every refused order contains a word the training grammar never uses (96–100% on Qwen, 97–100% on LFM2), against 81% of the orders answered. An order with a greeting, *please* or *thanks* (289 of the test orders) is refused 6.1% of the time by Qwen WORDS against 2.6% for other orders. The strict rule meets politeness the training grammar does not contain.

Corrections are learned, and best by WORDS. On test orders with held-out cue phrases, exact match is 62.9–75.3% on Qwen and 49.2–62.6% on LFM2, and the correction itself is applied in 80–88% and 72–78% of cases. WORDS is the best format on both models, separating from every other

Format	Neg. handled ↑	Real refused ↓	Corr. EM ↑	Corr. applied	Probes /40
Qwen3.5-0.8B					
EXR	99.0	3.3	74.5 ±4.3	86.5	36.0
JSON	98.0	3.2	70.0 ±2.8	85.7	37.8
EXR-SHORT	96.1	4.2	62.9 ±6.9	82.8	35.4
TAGGED	95.9	2.1	73.0 ±4.5	87.7	37.2
DSL	98.9	5.4	71.4 ±4.8	80.4	37.8
WORDS	98.6	3.4	**75.3** ±4.7	85.0	38.4
LFM2.5-350M					
EXR	98.2	7.8	55.5 ±4.1	77.0	36.0
JSON	97.6	10.1	49.2 ±10.7	72.1	34.2
EXR-SHORT	98.1	10.1	53.4 ±4.3	76.1	35.0
TAGGED	97.2	6.3	52.4 ±8.6	77.8	33.6
DSL	98.3	9.2	54.6 ±4.0	73.3	36.4
WORDS	98.2	6.7	**62.6** ±3.0	77.7	36.2

Table 4: Guardrail at the full pool, percent, mean of 5 seeds. Neg. handled: held-out negatives refused (614 per run). Real refused: test orders refused (D_{refuse}). Corr.: 600 test orders with held-out correction cues; applied: the corrected attribute is right. Probes: hand-written corrections correct, of 40. Bold/underline as in Table 2.

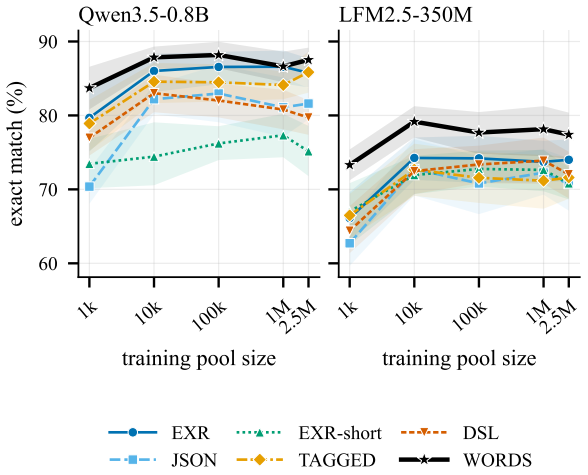


Figure 5: Exact match by training pool size, 95% bootstrap bands.

format on LFM2. The reported cheese failure is fixed in 60 of 60 full-pool runs. Training cues score 5–9 points above held-out cues on Qwen, so part of what is learned is the phrasing of the cue.

6.5 Pool size

WORDS is the most accurate format at every pool size on both models (Figure 5). Its lead over EXR is largest at 1k (+4.0 points on Qwen, +7.2 on LFM2) and narrows with pool size; its lead over DSL does not narrow on Qwen (+6.7 at 1k, +7.8 at the full pool). Natural words help most when data is scarce, but the advantage over a symbolic form of the same length does not go away.

6.6 Latency

Median latency at batch size one is linear in output tokens (Qwen: 29 ms + 26.5 ms per token, R^2=1.00; Appendix Figure 6) with throughput flat across formats, so WORDS answers 3.7× faster than EXR on Qwen and 4.4× on LFM2. The process's GPU memory moves only between 1,704 and 1,806 MiB on Qwen: format buys latency and accuracy, not memory.

7 Discussion

Which metric, when. If a parser's output is acted on without review, S is the number that matters: a format with higher A but higher S ships more wrong orders. A is the right number to compare with prior work, and $D_{\text{parse}} + D_{\text{refuse}}$ is what a retry or a human fallback has to absorb. On PIZZA the three point the same way for WORDS, which need not hold elsewhere; reporting all four outcomes makes the trade visible.

What transfers. The recipe is to compress the structure of the target and keep its vocabulary natural. It costs nothing at inference, needs no new labels, and is measurable on any parser with the four-way split. Whether the gap persists for larger models and other tasks is open (Limitations).

Strict refusal. Refusing anything that is not purely an order keeps the consumer from trusting half a ticket, at the cost of 2.1–10.1% of real orders. A softer rule, or negatives written in human rather than grammar style, could lower that cost; we leave it to future work and report the strict rule's price as is.

8 Conclusion

Output format is a free design variable for a fine-tuned parser. On PIZZA, with two model families, five pool sizes and five seeds, writing the target in plain words matches or beats the native format's accuracy with a quarter of the tokens and roughly halves the errors nobody can detect, mostly by getting negation right. The gain survives a refusal and self-correction guardrail trained into every run, and a tokenizer under which the word form is the longer one.

Limitations

One dataset. All results are on PIZZA, whose training pool comes from a grammar with a small vocabulary; the human-written test set is the main guard against learning the grammar alone. *Small models.* Both models are under a billion parameters; the effect may shrink with scale. *Author-generated guardrail data.* Negatives, corrections and probes are generated or written by us, and the human test set contains neither; 32 test orders (2.4%) use *make it*, *make that* or *hold on* in PIZZA's own sense, which our generator teaches as correction cues, bounding any effect on test exact match at 2.4 points. Every run includes the guardrail, so this campaign does not measure what the guardrail costs in accuracy. *Tuning.* Three learning rates were at the edge of the grid (Appendix Table 8); rank was not tuned; 8 of 300 runs (all LFM2) reached their best checkpoint within 100 steps of the step limit. *Generation.* 19 of 511,500 generations hit the token cap, all runaway outputs longer than any correct ticket. *Deployment.* The exported LFM2 GGUF agrees with its trained run on 93% of orders; no number in this paper comes from a GGUF.

Ethics Statement

PIZZA is released under CC-BY-NC-4.0 and contains no personal data; we redistribute no dataset text. All runs used two NVIDIA RTX 6000 Ada GPUs. shifts work to staff or customers; we report that rate rather than hide it.

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A Additional Results

B Integrity Checks

Before reading any result we checked: all 300 runs complete; exact match and parse validity recomputed from the raw outputs with the dataset's matcher equal the stored values in every run; negatives and corrections identical across models, formats and sizes for a seed; no evaluation utterance in any training subset; batched and batch-one generation agree on all but at most 2 of 200 utterances per run; left padding in every run. The analysis code and the checks are in the supplementary material.

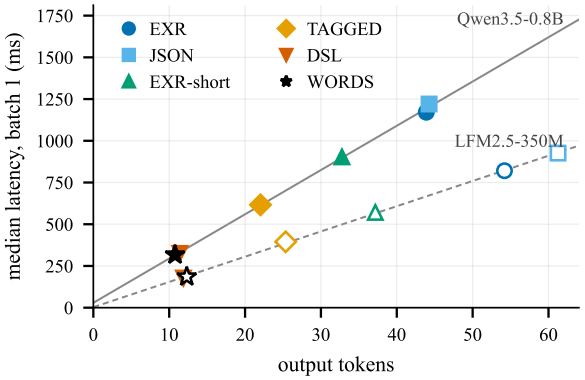


Figure 6: Median latency at batch size one against output tokens, mean over five seeds; filled: Qwen, hollow: LFM2. Lines: least-squares fits.

C Deployment Demonstration

The café parser that motivated this work, a separate model with its own compact format, runs entirely on an iPad (iPad15,7, A16) through llama.cpp with Metal offload: Q8_0 weights (795 MiB file, 350 MiB resident), 2.5 s load time, median 167 ms and p90 354 ms per order, 16 tok/s. Latency on the device is close to linear in output tokens, so the token savings of a compact format translate directly into response time. This is a demonstration on ten orders, not an evaluation.

	1k		10k		100k		1M		2.5M	
	A	S	A	S	A	S	A	S	A	S
Qwen3.5-0.8B										
EXR	79.7±4.9	17.0	86.0±2.7	9.6	86.6±2.0	9.2	86.6±2.1	10.1	85.8±1.6	10.3
JSON	70.4±1.4	25.1	82.2±1.3	14.6	83.0±4.4	12.9	81.1±4.6	11.5	81.6±1.3	14.9
EXR-SHORT	73.5±3.5	18.8	74.4±4.9	15.0	76.2±1.4	15.1	77.3±2.9	13.4	75.2±3.6	17.4
TAGGED	78.9±3.7	16.8	84.6±2.9	12.1	84.5±2.5	10.8	84.1±1.1	12.4	85.9±2.9	11.6
DSL	77.0±1.9	10.5	83.0±2.4	8.6	82.0±1.9	8.9	80.8±2.6	8.1	79.7±1.9	8.3
WORDS	83.7±3.1	5.6	87.9±0.4	4.6	88.2±1.3	4.8	86.6±1.8	5.1	87.5±0.8	4.5
LFM2.5-350M										
EXR	66.1±3.6	25.9	74.3±2.5	18.6	74.2±3.3	17.4	73.7±2.4	17.4	74.0±2.5	17.9
JSON	62.7±3.0	29.4	72.7±4.1	16.4	70.8±4.3	18.3	72.3±2.9	18.1	70.7±3.6	19.1
EXR-SHORT	67.0±4.6	22.8	71.9±1.9	18.6	72.8±1.6	19.1	72.6±2.4	17.6	70.8±0.7	16.9
TAGGED	66.5±2.7	24.0	72.7±3.6	20.5	71.6±3.5	18.8	71.2±3.8	18.9	71.6±2.5	21.4
DSL	64.4±2.9	19.2	72.5±2.8	15.2	73.4±1.8	14.2	73.9±2.9	13.6	72.0±3.0	13.1
WORDS	73.3±0.9	14.7	79.2±1.3	10.1	77.7±2.7	9.7	78.1±3.5	8.7	77.4±3.2	10.0

Table 5: Exact match A (± seed SD) and silent rate S at every pool size, percent.

Difference (points)	1k	10k	100k	1M	2.5M
Qwen3.5-0.8B					
WORDS−DSL	+6.7 [+3.4, +10.2]†	+4.8 [+2.3, +7.3]†	+6.2 [+3.9, +8.4]†	+5.8 [+3.4, +8.6]†	+7.8 [+6.1, +9.5]†
EXR−JSON	+9.3 [+5.4, +12.8]†	+3.8 [+0.2, +6.7]	+3.6 [-1.1, +7.8]	+5.5 [+2.2, +9.2]†	+4.2 [+2.4, +5.8]†
EXR−EXR-SHORT	+6.2 [+3.3, +9.2]†	+11.6 [+4.7, +16.6]†	+10.3 [+7.9, +12.7]†	+9.3 [+6.2, +12.5]†	+10.6 [+7.9, +13.4]†
WORDS−EXR	+4.0 [+0.5, +7.7]	+1.8 [-0.4, +4.4]	+1.6 [-0.8, +4.2]	+0.0 [-1.3, +1.3]	+1.8 [+0.1, +3.4]
S: EXR−WORDS	+11.4 [+8.9, +13.9]†	+5.1 [+3.3, +7.1]†	+4.4 [+2.8, +6.2]†	+5.0 [+3.1, +7.1]†	+5.8 [+4.3, +7.4]†
LFM2.5-350M					
WORDS−DSL	+8.9 [+5.5, +11.9]†	+6.7 [+3.5, +9.9]†	+4.3 [+2.2, +6.3]†	+4.3 [-1.2, +8.9]	+5.4 [+3.3, +7.4]†
EXR−JSON	+3.4 [-1.7, +8.5]	+1.5 [-3.1, +5.8]	+3.4 [-0.5, +9.0]	+1.5 [-1.4, +4.1]	+3.3 [-0.4, +7.3]
EXR−EXR-SHORT	-0.9 [-4.8, +2.2]	+2.3 [-0.8, +5.5]	+1.4 [-2.2, +4.3]	+1.1 [-2.0, +5.1]	+3.2 [+0.9, +5.4]†
WORDS−EXR	+7.2 [+3.7, +10.7]†	+4.9 [+2.5, +7.5]†	+3.5 [-0.0, +7.5]	+4.4 [+1.0, +7.8]†	+3.4 [-0.9, +7.6]
S: EXR−WORDS	+11.2 [+8.6, +13.7]†	+8.6 [+5.9, +11.4]†	+7.7 [+4.3, +10.6]†	+8.7 [+6.0, +11.5]†	+7.9 [+4.2, +11.6]†

Table 6: Table 3 with 95% hierarchical bootstrap intervals.

Format	Chars	Qwen3.5		LFM2.5	
		tok.	ch./tok.	tok.	ch./tok.
EXR	123.0	45.0	2.7	58.0	2.1
JSON	150.7	45.6	3.3	64.7	2.3
EXR-SHORT	66.0	33.4	2.0	39.0	1.7
TAGGED	63.4	21.7	2.9	25.7	2.5
DSL	34.9	10.8	3.2	11.7	3.0
WORDS	43.1	10.3	4.2	11.9	3.6

Table 7: Mean characters and tokens of the gold targets over all 1,357 test orders, for both tokenizers.

Format	1e-4	2e-4	5e-4	
Qwen3.5-0.8B				
EXR	84.77	**85.82**	80.36	
JSON	**83.62**	83.62	80.27	edge
EXR-SHORT	76.82	79.89	**79.98**	edge
TAGGED	86.11	**87.07**	81.13	
DSL	81.13	**82.95**	79.02	
WORDS	86.11	**86.78**	79.50	
LFM2.5-350M				
EXR	69.83	**75.57**	70.50	
JSON	69.54	**70.02**	69.54	
EXR-SHORT	71.46	**75.77**	66.86	
TAGGED	74.62	**75.00**	70.21	
DSL	**75.57**	72.70	67.05	edge
WORDS	78.93	**79.02**	69.83	

Table 8: Learning-rate selection: dev exact match at a 100k pool, mean of 3 seeds; bold is the rate used. "edge": chosen at the end of the grid.

Format	junk	long	non-order	off-menu	mixed	unknown word	probe
Qwen3.5-0.8B							
EXR	100.0	100.0	99.4	99.8	99.3	96.1	100.0
JSON	100.0	100.0	100.0	98.8	98.3	92.9	100.0
EXR-SHORT	99.7	100.0	99.4	97.5	96.3	87.0	100.0
TAGGED	100.0	100.0	99.1	95.9	96.9	86.3	100.0
DSL	100.0	100.0	99.4	99.8	99.0	96.0	100.0
WORDS	100.0	100.0	99.7	99.3	97.6	95.9	100.0
LFM2.5-350M							
EXR	100.0	100.0	100.0	98.9	99.4	92.2	100.0
JSON	100.0	100.0	100.0	98.3	99.7	89.6	100.0
EXR-SHORT	100.0	100.0	99.7	98.7	99.8	91.6	100.0
TAGGED	100.0	100.0	100.0	96.5	98.3	90.5	100.0
DSL	100.0	100.0	100.0	98.3	99.3	93.7	100.0
WORDS	100.0	100.0	100.0	98.4	99.2	93.0	100.0

Table 9: Held-out negatives refused, by kind, full pool, percent.

Format	swap	retract	only	size	number	drink	cancel	restart	decoy
Qwen3.5-0.8B									
EXR	73.1	73.2	82.3	81.7	86.7	59.6	61.8	78.6	74.4
JSON	66.8	63.5	85.6	75.7	80.4	56.9	66.0	77.8	64.9
EXR-SHORT	60.3	58.0	79.7	72.6	74.9	46.3	57.6	61.0	59.6
TAGGED	72.3	67.2	84.7	75.2	82.7	63.8	67.6	79.2	70.1
DSL	72.9	72.1	80.3	72.1	78.9	62.8	67.8	75.6	63.3
WORDS	72.3	74.9	89.1	81.1	79.7	60.1	68.2	77.9	73.7
LFM2.5-350M									
EXR	53.2	53.9	71.3	67.3	73.3	32.5	44.4	52.6	51.4
JSON	49.8	50.9	65.6	60.2	63.5	24.3	39.4	37.1	45.8
EXR-SHORT	54.7	56.2	69.4	64.8	69.6	28.3	37.9	44.5	49.9
TAGGED	45.6	54.7	72.6	56.9	62.9	34.9	40.3	50.9	51.2
DSL	54.0	59.8	66.8	58.3	66.7	31.8	42.6	54.4	51.2
WORDS	60.0	65.7	70.9	76.2	70.9	43.1	51.6	53.8	63.4

Table 10: Held-out corrections, exact match by kind, full pool, percent.

Format	n	line count	negation	topping swapped	value swapped	invented	dropped	other structure
Qwen3.5-0.8B								
EXR	698	17	19	15	31	9	9	1
JSON	1011	10	34	11	28	3	13	1
EXR-SHORT	1180	18	19	18	18	8	18	2
TAGGED	786	14	33	11	21	4	14	4
DSL	561	15	30	12	20	7	13	3
WORDS	303	36	5	3	15	11	25	5
LFM2.5-350M								
EXR	1216	30	20	7	15	4	22	1
JSON	1298	28	14	7	18	7	25	1
EXR-SHORT	1150	27	22	9	17	7	16	2
TAGGED	1450	23	24	6	20	2	23	1
DSL	887	27	28	7	12	6	14	6
WORDS	681	35	6	4	9	19	24	4

Table 11: Silent errors by kind, full pool, all seeds pooled: n errors, and the percent of them in each kind.